

South China Sea model validation

Completed regional review for the selected 2000s northern South China Sea model. All 374 taxa were reviewed; 36 supported group corrections are adopted regionally. The linked appendix records confidence and assumptions. Estimates remain provisional.

Field	Entry
Region	South China Sea LME_036 Regional workbook
Catch source	Sea Around Us; 1950–2019; source contains landings, discards and total catch. Region Catch ZIP Taxon catch Snapshot documented 3 Sep 2026; Assessment: 2019 landings, as recorded in Overview / Settings. Total catch includes landings + discards.
Selected article	Cheung, Wai Lung (2007). Vulnerability of marine fishes to fishing: from global overview to the northern South China Sea. UBC doctoral thesis. DOI 10.14288/1.0074894 Thesis PDF Paper folder
Other known articles	• Pauly & Christensen (1993). Stratified models of large marine ecosystems: application to the South China Sea. Local record • Pitcher et al. (2000). Marine Reserves and the Restoration of Fisheries and Marine Ecosystems in the South China Sea. Local record • Lin, Chen & Lin (2020). Trophic model of a deep-sea ecosystem with methane seeps in the South China Sea. Local record DOI • Zhang et al. (2022). Effects of ocean warming and fishing on the coral reef ecosystem: A case study of Xisha Islands, South China Sea. Local record DOI • Jiang et al. (2023). A preliminary model of the mangrove ecosystem of Dongzhaigang Bay, Hainan, (China) based on Ecopath and Ecospace. Local record DOI • Feng et al. (2026). Ecosystem characteristics and ecological carrying capacity of a subtropical marine ranch in the pearl river estuary: an ecopath modeling approach. Local record DOI
Selected model	36_1_South_China_Sea_SCS-2007_Northern_South_China_Sea_(2000s) Selected, 2000s northern shelf; 38 source groups. Folder JSON Saved runtime .
Other models from the same article	Northern South China Sea (1970s): same northern shelf and 38 source groups; earlier biomass, mortality and catch parameterization with adult-heavy stages. Kept separately from the selected 2000s model.
Selection rationale	Article – ? Model – newer model (2000 vs 1970)
Model extraction	source reconstruction report - several internal conflicts within the paper. Full conflict report. - Diet Sums – only rounding errors. max error = 0.0011, group 33 (Pelagic sharks and rays). - GS not reported; defaulted to 0.2 for all consumer groups, 0 for producers/detritus/import groups - BA not reported; auto completed through LIM - state comparison.

Field	Entry
SPPR calculation	Removed groups from calculation: - seabirds (TE<0.1%, TL=3.4, sppr_TE=3.75M, sppr_GE=21K) - other mammals (TE<0.1%, TL=3.9, sppr_TE=800K, sppr_GE=13K) - Pinnipeds (TE<0.1%, TL=4.1, sppr_TE=340K, sppr_GE=3400) - Marine Turtles (TE<1%, TL=3, sppr_TE=20K, sppr_GE=724)

Field	Entry
GE diagnostics	WARN: near-zero efficiency in seabirds (34). No negative SPPR across basal-source columns. rho_living = 0.3209187537609611; b = 0.00408138212364339. Detritus (38): SPPR = 1.007782936216668. Strict model and SPPR balance flags are true.
TE diagnostics	WARN: near-zero efficiency in seabirds, other mammals and pinnipeds (34–36). No negative SPPR across basal-source columns. rho_living = 0.6163296517365333. Detritus (38): SPPR = 0.995222499887902. Strict model and SPPR balance flags are true.
Geographic fit	A. Region covered by study: approximately 15–25%. B. Study area covered by region: approximately 90–100%.
Temporal fit	Model period: 2000s (no verified single model year). Fixed coefficients and stage weights are applied to catch years 1950–2019, including the 2019 reference. They do not reconstruct annual ecosystems and transfer a northern-shelf model to whole-LME catch.
Other	- The accepted saved and reproduced outputs retain groups 34–37 (seabirds, other mammals, pinnipeds and turtles), all with finite direct SPPR. The researcher removal note describes a different configuration. With Egestion returns OK. All three complete source matrices have no negative or nonfinite entries, including unfished groups.
Open issues and next action	- Test how much SPPR and PPR values change when using the 1970 model. - Add article selection reason when we review other articles
Review and reproducibility	Researcher name: Ido Carmel review date: 30/09/2026

Taxon mapping and coverage

Reference: 2019 landings, excluding discards. All sources, all taxa and groups;

There are 38 source groups plus one computational import and 374 catch taxa.

Simple-chain PPR totals 837,452,937.79 t C in 2019.

The 34 taxa that lack an assigned trophic level have zero landings that year and therefore contribute zero PPR.

Simple-chain PPR is calculated with the simple trophic chain method.

[Excel taxon appendix and Sources](#) contains all taxa, sorted by unrounded PPR, and allocation values and assumptions.

Overall confidence	Taxa	Catch percentage	PPR percentage
High	142	39.0208%	31.8829%
Medium	165	29.8334%	47.7207%
Low	57	18.9426%	16.3853%
Very low	10	12.2032%	4.0110%
Unresolved	0	0.0000%	0.0000%

The summary uses the weaker required membership or allocation confidence. The following tables show each component separately.

Group assignment rules

Plain-language rule	Confidence	PPR percentage
Documented taxonomy and ecology unambiguously meet the group definition	High	48.1160%
Ecological fit is partial or the source definition is conflicting	Low	16.3853%
The source explicitly assigns the taxon to the model group	High	15.2077%
Taxonomy, ecology and group definitions support the eligible group set	Medium	15.1324%
A broad catch label is approximated by compatible model groups	Very low	4.0110%
Documented size and habitat support an ecological assignment	Medium	1.1476%
The closest represented ecological analogue substitutes for an absent group	Very low	0.0000%

Membership sources: [Cheung group definitions](#); [adopted regional mapping and evidence](#). Taxonomic authorities and ecological limitations are described on the Excel Sources sheet.

Allocation and mapping uncertainty

Plain-language rule	Confidence	PPR percentage
Model catch proportions supply an assumed allocation across eligible groups	Medium	65.0799%
The taxon is assigned to one group and needs no split	High	34.9201%

Each percentage is the simple-chain PPR associated with taxa using that rule.

All 374 catch taxa have an assigned group: 100% of 2019 landings and simple-chain PPR are mapped. Unresolved taxa: none. This includes 10 Very low assignments covering 12.2032% of landings and 4.0110% of simple-chain PPR; their taxa, weights and reasons are flagged in the Excel appendix.

Very low decisions

Affected taxa	Why confidence is very low
Marine fishes not identified; Perciformes; Marine finfishes not identified; Chondrichthyes; Miscellaneous aquatic invertebrates; Marine groundfishes not identified; Marine pelagic fishes not identified; Gadiformes	Catch composition is unknown; compatible named and residual pools provide a broad approximation. Exact candidates and assumptions are in the appendix.
Lampris guttatus	The historical opah label represents offshore deep-living fishes and a subsequently divided species complex. The adopted shelf large-pelagic pair remains a closest represented analogue without an established link to the source guild's sampled assemblage.
Lepidocybium flavobrunneum	Escolar is principally a continental-slope fish below 200 m with upward night movements. The adopted shelf large-pelagic pair remains a weak analogue; a competing benthopelagic interpretation and the missing deep-water compartment are unresolved.

Whole model-pool catch proportions are assumed representative of regional taxon catches across 1950–2019 and all catch bases. These are not observed target-catch mixtures. Another 57 taxa retain Low confidence because of size, habitat or source-group ambiguity. The four residual fish labels follow verified bony-fish reporting scope; dedicated demersal and pelagic shark/ray pools are excluded. Unknown composition and whole-region transfer remain Very low; complete source catch supplies Medium allocation weights. [Reporting scope evidence](#)

Allocation sources: [selected model catch and biomass fields](#); [adopted candidate decisions and calculations](#). Prior stage assumptions and their source conflicts remain in the regional Allocation assumptions ledger.

High mapping coverage does not establish model validity. The northern shelf model is extrapolated to the wider region; source-group ambiguity and the diagnostic WARN conditions above remain.

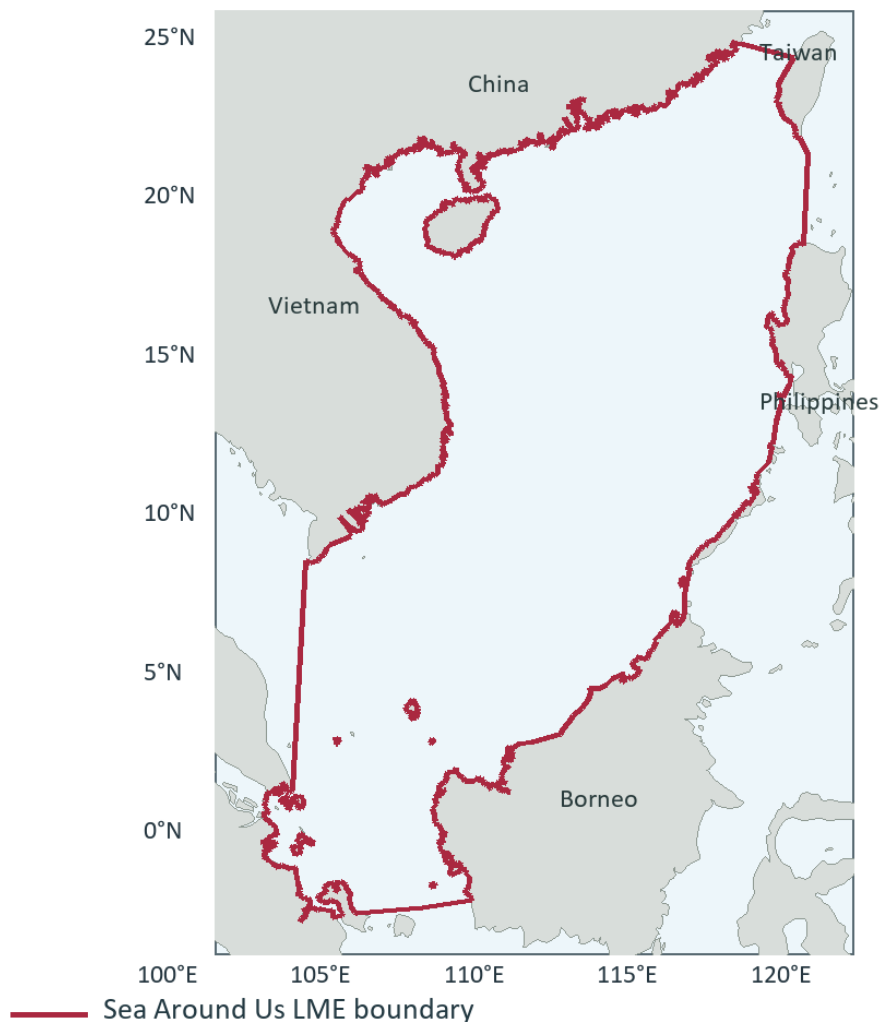
Geographic evidence screenshots

R is LME036; S is the northern shelf coast-to-broken-line area in Cheung Figure 6.1. The grid and shared coastal landmarks support the approximate ranges above.

Region area

Target region boundary

LME_036 South China Sea



Natural Earth land context. Generated from retained GeoJSON.

Display only; no numerical overlap inferred.

LME_036 South China Sea (R). Retained researcher figure drawn from [Sea Around Us boundary](#); dataset release/date not recorded. [Source and display method](#) | [Image](#).

Study area in the article

Study boundary for the selected model

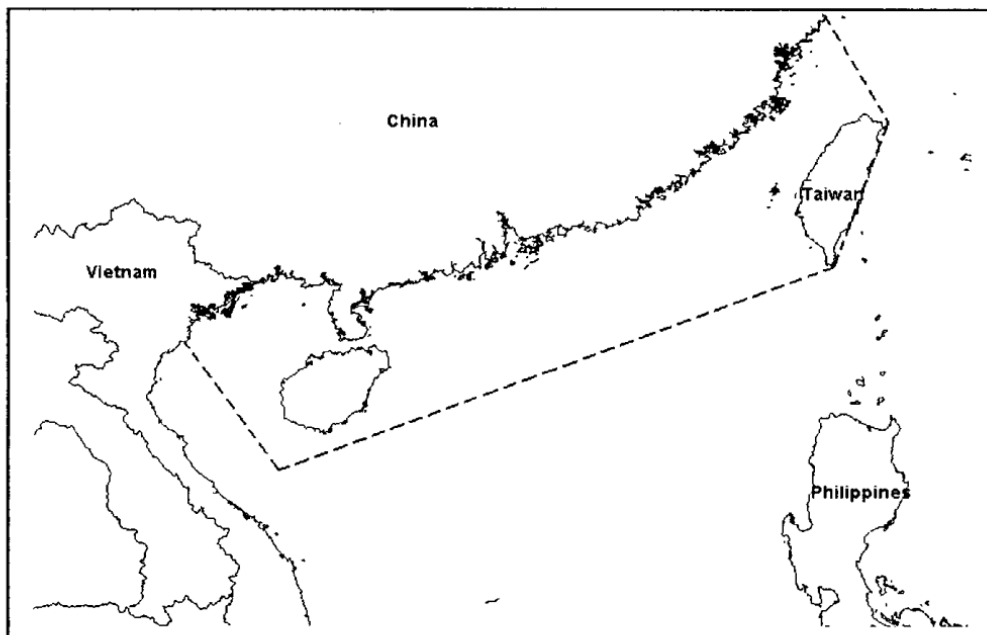


Figure 6.1. Map of the northern South China Sea, emphasizing the study area (from the coast to the broken line). As fishers from the PRC and Vietnam traditionally fish in the Gulf of Tonkin, the PRC and Vietnam governments agreed to establish common fishing zone in the Gulf of Tonkin, at least for a limited period (Vietnam-China Tonkin Gulf Fishing Co-operation Agreement).

Cheung (2007), Figure 6.1, printed p.170 / PDF p.185. Northern South China Sea shelf (S), 2000s selected model; coast to broken line. [Source PDF](#) | [DOI](#) | [Image](#).

Visual estimates transfer geographic landmarks between maps; raw screenshot pixel areas were not compared. Boundary interpretation near Taiwan and differing map scales limit precision.